ICS 253 – Discrete Structures
Chapter 6: Induction and Recursion

Question bank: 90 multiple choice + 10 written problems

Questions from ICS 253 old quizzes and homework (AI_Work). **[new]** marks new questions written in the same style.

Part A: Multiple choice

Q1. [new]

Let $S(n)$ be a statement about a positive integer n. In an inductive proof that $S(n)$ is true for every positive integer n, what must be proven in the inductive step?

(A) $S(1)$ is true.

(B) For every $k \geq 1$, $S(k)$ implies $S(k+1)$.

(C) $S(k)$ is true for some positive integer k.

(D) For every $k \geq 1$, $S(k+1)$ implies $S(k)$.

Answer: (B). The base case proves $S(1)$; the inductive step proves the implication $S(k) \to S(k+1)$ for every $k \geq 1$.

My answer

Q2. Quiz 3 (Term 232)

The rule of inference $\big(P(1) \wedge \forall k\,(P(k) \to P(k+1))\big) \to \forall n\,P(n)$ represents:

(A) proof by induction

(B) proof by strong induction

(C) direct proof

(D) indirect proof

Answer: (A). A base case $P(1)$ plus the step $P(k) \to P(k+1)$ is exactly the principle of (ordinary) mathematical induction.

My answer

Q3. Quiz 3 Sec 3 (Term 232)

What is the purpose of the base case in a proof by mathematical induction?

(A) It establishes the truth of the statement for all n.

(B) It simplifies the induction hypothesis.

(C) It ensures the statement holds for infinitely many values of n.

(D) It verifies the statement for some specific initial value of n.

Answer: (D). The base case checks the first value(s) directly; the inductive step then carries the truth forward.

My answer

Q4. [new]

In the inductive step of a proof, the writer says: "Suppose that for a positive integer k, $P(k)$ is true. We will show that $P(k+1)$ is true." The supposition that $P(k)$ is true is called the:

(A) base case

(B) well-ordering principle

(C) inductive hypothesis

(D) exclusion statement

Answer: (C). The assumption $P(k)$ used to derive $P(k+1)$ is the inductive hypothesis.

My answer

Q5. Quiz 3 (Term 232)

When using mathematical induction to prove that $1^2 + 2^2 + \cdots + n^2 = \dfrac{n(n+1)(2n+1)}{6}$, what do we show in the basis step?

(A) $\sum_{k=1}^{n} k^2 = \dfrac{n(n+1)(n+2)}{6}$ for some k

(B) $P(k) \to P(k+1)$

(C) $1^2 = \dfrac{(1)(2)(3)}{6}$

(D) $P(k+1)$

Answer: (C). Substitute $n = 1$: the left side is 1^2 and the right side is $\frac{1\cdot 2\cdot 3}{6} = 1$.

My answer

Q6. Quiz 3 Alkhulaifi

When using mathematical induction to prove that $1^3 + 2^3 + \cdots + n^3 = \dfrac{n^2(n+1)^2}{4}$ for all integers $n \geq 1$, the basis step is to show that

(A) $1^3 + 2^3 + \cdots + n^3 = \dfrac{n^2(n+1)^2}{4}$

(B) $1^3 + 2^3 = \dfrac{2^2 \cdot 3^2}{4}$

(C) $1^3 + 2^3 + \cdots + (k+1)^3 = \dfrac{(k+1)^2(k+2)^2}{4}$

(D) $1^3 = \dfrac{1^2 \cdot 2^2}{4}$

Answer: (D). The smallest value is $n = 1$, so the basis step checks $1^3 = \frac{1^2 \cdot 2^2}{4} = 1$.

My answer

Q7. [new]

Let $P(n)$ be the statement $1 + 3 + 5 + \cdots + (2n-1) = n^2$. Which of the following is $P(k+1)$?

(A) $1 + 3 + 5 + \cdots + (2k-1) + (2k+1) = (k+1)^2$

(B) $1 + 3 + 5 + \cdots + (2k-1) = (k+1)^2$

(C) $1 + 3 + 5 + \cdots + (2k+1) = k^2 + 1$

(D) $1 + 3 + 5 + \cdots + (2k+3) = (k+1)^2$

Answer: (A). Replace n by $k+1$: the last term becomes $2(k+1) - 1 = 2k + 1$ and the right side becomes $(k+1)^2$.

My answer

Q8. [new]

In the inductive step of the proof that $\sum_{j=1}^{n} j = \dfrac{n(n+1)}{2}$, the left side of $P(k+1)$ is written as $\left(\sum_{j=1}^{k} j\right) + (k+1)$. After applying the inductive hypothesis, this expression becomes:

(A) $\dfrac{(k+1)(k+2)}{2} + (k+1)$

(B) $\dfrac{k(k+1)}{2} + (k+1)$

(C) $\dfrac{k(k+1)}{2} + k$

(D) $\dfrac{k(k-1)}{2} + (k+1)$

Answer: (B). The inductive hypothesis replaces $\sum_{j=1}^{k} j$ by $\frac{k(k+1)}{2}$; then $\frac{k(k+1)}{2} + (k+1) = \frac{(k+1)(k+2)}{2}$.

My answer

Q9. [new]

Using the identity $\sum_{j=1}^{n} j^2 = \dfrac{n(n+1)(2n+1)}{6}$, the value of $1^2 + 2^2 + \cdots + 10^2$ is:

(A) 285

(B) 330

(C) 506

(D) 385

Answer: (D). $\frac{10 \cdot 11 \cdot 21}{6} = 385$.

My answer

Q10. [new]

To prove by induction that $\sum_{j=1}^{n} j \cdot j! = (n+1)! - 1$ for every positive integer n, what must be proven in the inductive step?

(A) For $k \geq 1$: if $\sum_{j=1}^{k} j \cdot j! = (k+1)! - 1$, then $\sum_{j=1}^{k+1} j \cdot j! = (k+2)! - 1$.

(B) For $k \geq 1$: if $\sum_{j=1}^{k} j \cdot j! = (k+1)! - 1$, then $\sum_{j=1}^{k+1} j \cdot j! = (k+1)! - 1$.

(C) $1 \cdot 1! = 2! - 1$.

(D) For $k \geq 1$: if $\sum_{j=1}^{k+1} j \cdot j! = (k+2)! - 1$, then $\sum_{j=1}^{k} j \cdot j! = (k+1)! - 1$.

Answer: (A). Assume $P(k)$ and prove $P(k+1)$, where the right side of $P(k+1)$ is $((k+1)+1)! - 1 = (k+2)! - 1$. (C) is the base case.

My answer

Q11. [new]

The value of $1 \cdot 1! + 2 \cdot 2! + 3 \cdot 3! + 4 \cdot 4!$ is:

(A) 33

(B) 119

(C) 120

(D) 719

Answer: (B). $1 + 4 + 18 + 96 = 119 = 5! - 1$, consistent with the identity $\sum_{j=1}^{n} j \cdot j! = (n+1)! - 1$.

My answer

Q12. [new]

The inequality $2^n > n^2$ is to be proven by induction for all integers $n \geq n_0$. What is the smallest n_0 for which the statement is true for every $n \geq n_0$ (and hence the value to use in the base case)?

(A) 3

(B) 4

(C) 5

(D) 6

Answer: (C). $2^3 = 8 < 9$, $2^4 = 16 = 16$ (not $>$), but $2^5 = 32 > 25$ and the inequality holds from then on.

My answer

Q13. [new]

To prove $n! > 2^n$ for all integers $n \geq 4$, the base case shows:

(A) $1! > 2^1$

(B) $4! > 2^4$, i.e. $24 > 16$

(C) $3! > 2^3$, i.e. $6 > 8$

(D) $(k+1)! > 2^{k+1}$

Answer: (B). The base case is the smallest value in the range, $n = 4$: $24 > 16$.

My answer

Q14. [new]

In the inductive step of the proof that $n! > 2^n$ for $n \geq 4$, the following chain is written for $k \geq 4$:

$$(k+1)! = (k+1) \cdot k! \overset{(1)}{>} (k+1) \cdot 2^k \overset{(2)}{>} 2 \cdot 2^k = 2^{k+1}.$$

What justifies inequality (1)?

(A) The inductive hypothesis $k! > 2^k$.

(B) The fact that $k + 1 > 2$.

(C) The base case $4! > 2^4$.

(D) The definition $(k+1)! = (k+1) \cdot k!$.

Answer: (A). Step (1) multiplies the inductive hypothesis $k! > 2^k$ by the positive number $k + 1$; step (2) uses $k + 1 > 2$.

My answer

Q15. Quiz 4 Alenzi

Claim: for all $n \geq 1$, $n! \leq 2^n$. "Proof": Base case $1! = 1 \leq 2$. Assume $k! \leq 2^k$. Then

$$(k+1)! = (k+1) \cdot k! \overset{(1)}{\leq} (k+1) \cdot 2^k \overset{(2)}{\leq} 2 \cdot 2^k = 2^{k+1}.$$

Which step is false and invalidates the proof?

(A) The base case $1! \leq 2^1$.

(B) The identity $(k+1)! = (k+1) \cdot k!$.

(C) Inequality (1), $(k+1) \cdot k! \leq (k+1) \cdot 2^k$.

(D) Inequality (2), which needs $k+1 \le 2$; this fails for every $k \ge 2$.

Answer: (D). (2) requires $k+1 \le 2$, false for $k \ge 2$, so the inductive step fails (indeed $4! = 24 > 16 = 2^4$).

My answer

Q16. [new]

It can be proven by induction that $\sum_{j=1}^{n} \frac{1}{j(j+1)} = \frac{n}{n+1}$. What is $\frac{1}{1\cdot 2} + \frac{1}{2\cdot 3} + \cdots + \frac{1}{9\cdot 10}$?

(A) $\frac{9}{10}$

(B) $\frac{10}{11}$

(C) $\frac{8}{9}$

(D) 1

Answer: (A). There are 9 terms ($j = 1$ to 9), so the sum is $\frac{9}{10}$.

My answer

Q17. [new]

A well-known false argument "proves" that in any set of $n \ge 1$ horses all horses have the same colour: remove one horse, apply the hypothesis to the remaining k horses, then remove a different horse and apply it again; the two groups overlap, so all $k+1$ horses share a colour. Where does the argument break?

(A) The base case $n = 1$ is false.

(B) The inductive step fails when going from $k = 1$ to $k+1 = 2$, because the two groups of one horse do not overlap.

(C) The inductive step fails for every $k \ge 1$.

(D) Induction cannot be used for statements about sets.

Answer: (B). For $k+1 = 2$ the two groups are $\{h_1\}$ and $\{h_2\}$, which share no horse, so the implication $P(1) \to P(2)$ is not established.

My answer

Q18. [new]

A student claims that $n^2 + n$ is odd for every positive integer n and "proves" it as follows: assume $k^2 + k$ is odd; then $(k+1)^2 + (k+1) = (k^2 + k) + 2(k+1)$ is odd plus even, hence odd. What is wrong?

(A) The algebra in the inductive step is incorrect.

(B) The inductive hypothesis is used incorrectly.

(C) The statement is true, so nothing is wrong.

(D) No base case was proven, and in fact $P(1)$ is false since $1^2 + 1 = 2$ is even.

Answer: (D). The inductive step is valid, but without a true base case nothing follows; here every $n^2 + n = n(n+1)$ is even.

My answer

Q19. [new]

Suppose $P(1)$ is true and, for every $k \geq 1$, $P(k)$ implies $P(k+2)$. What can be concluded?

(A) $P(n)$ is true for every positive integer n.

(B) $P(n)$ is true for every even positive integer n.

(C) $P(n)$ is true for every odd positive integer n.

(D) Nothing beyond $P(1)$.

Answer: (C). $P(1) \Rightarrow P(3) \Rightarrow P(5) \Rightarrow \cdots$; nothing is said about $P(2)$.

My answer

Q20. [new]

In the inductive step of a proof that $3^n \geq 2n + 1$ for all $n \geq 0$, the following lines appear for $k \geq 0$:

$$3^{k+1} \overset{(1)}{=} 3 \cdot 3^k \overset{(2)}{\geq} 3(2k+1) \overset{(3)}{=} 6k + 3 \overset{(4)}{\geq} 2k + 3 = 2(k+1) + 1.$$

In which line is the inductive hypothesis used?

(A) (1)

(B) (2)

(C) (3)

(D) (4)

Answer: (B). Line (2) replaces 3^k by the smaller quantity $2k + 1$, which is the inductive hypothesis $3^k \geq 2k + 1$. Line (4) uses $k \geq 0$.

My answer

Q21. [new]

To prove that $P(n)$ holds for every integer $n \geq 5$ by induction, one should prove:

(A) $P(1)$, and for all $k \geq 1$, $P(k) \to P(k+1)$.

(B) $P(5)$, and for all $k \geq 5$, $P(k) \to P(k+1)$.

(C) $P(5)$, and for all $k \geq 1$, $P(k+1) \to P(k)$.

(D) $P(5)$ and $P(6)$ only.

Answer: (B). The base case is the first value $n = 5$, and the inductive step is proved for every $k \geq 5$.

My answer

Q22. [new]

An inductive proof that 3 evenly divides $4^n - 1$ for every positive integer n uses the inductive hypothesis "3 evenly divides $4^k - 1$". Which statement is equivalent to this hypothesis?

(A) $4^k = 3m$ for some integer m

(B) $4^k - 1 = 3k$

(C) $4^k = 3m + 1$ for some integer m

(D) $4^{k+1} - 1 = 3m$ for some integer m

Answer: (C). $4^k - 1 = 3m$ for some integer m, i.e. $4^k = 3m + 1$.

My answer

Q23. [new]

Continuing the proof that $3 \mid 4^n - 1$: with $4^k = 3m + 1$, which expression shows that 3 evenly divides $4^{k+1} - 1$?

(A) $4^{k+1} - 1 = 4(3m + 1) - 1 = 3(4m + 1)$

(B) $4^{k+1} - 1 = 3m + 4$

(C) $4^{k+1} - 1 = 4 \cdot 3m$

(D) $4^{k+1} - 1 = (3m + 1)^2 - 1$

Answer: (A). $4^{k+1} - 1 = 4 \cdot 4^k - 1 = 12m + 3 = 3(4m + 1)$, and $4m + 1$ is an integer.

My answer

Q24. [new]

In an inductive proof that 3 evenly divides $n^3 + 2n$ for every positive integer n, which identity is used in the inductive step?

(A) $(k + 1)^3 + 2(k + 1) = (k^3 + 2k) + 3$

(B) $(k + 1)^3 + 2(k + 1) = 3(k^3 + 2k)$

(C) $(k + 1)^3 + 2(k + 1) = (k^3 + 2k) + 3k^2 + 1$

(D) $(k + 1)^3 + 2(k + 1) = (k^3 + 2k) + 3(k^2 + k + 1)$

Answer: (D). $(k + 1)^3 + 2(k + 1) = k^3 + 3k^2 + 3k + 1 + 2k + 2 = (k^3 + 2k) + 3(k^2 + k + 1)$; both parts are multiples of 3.

My answer

Q25. [new]

To prove that 7 evenly divides $9^n - 2^n$ for every positive integer n, the inductive step rewrites $9^{k+1} - 2^{k+1}$ as:

(A) $9(9^k - 2^k) + 7 \cdot 2^k$

(B) $9(9^k - 2^k) - 7 \cdot 2^k$

(C) $2(9^k - 2^k) + 9 \cdot 2^k$

(D) $7(9^k - 2^k)$

Answer: (A). $9 \cdot 9^k - 9 \cdot 2^k + 9 \cdot 2^k - 2 \cdot 2^k = 9(9^k - 2^k) + 7 \cdot 2^k$; both terms are divisible by 7.

My answer

Q26. [new]

Define the sequence $a_1 = 1$ and $a_n = a_{n-1} + 2n - 1$ for $n \geq 2$. Which explicit formula can be proven by induction?

(A) $a_n = 2n - 1$

(B) $a_n = n^2$

(C) $a_n = n^2 - n + 1$

(D) $a_n = 2^n - 1$

Answer: (B). $a_1 = 1, a_2 = 4, a_3 = 9, \ldots$; and if $a_k = k^2$ then $a_{k+1} = k^2 + 2(k+1) - 1 = (k+1)^2$.

My answer

Q27. [new]

A sequence is defined by $a_0 = 1$ and $a_n = 3a_{n-1} + 2$ for $n \geq 1$. Which statement is equivalent to this recurrence relation (useful in an inductive step that goes from k to $k+1$)?

(A) $a_{k+1} = 3a_k + 2$ for $k \geq 1$

(B) $a_k = 3a_{k+1} + 2$ for $k \geq 0$

(C) $a_{k+1} = 3a_k + 2$ for $k \geq 0$

(D) $a_{k+1} = 3a_{k-1} + 2$ for $k \geq 1$

Answer: (C). Substituting $n = k + 1$, the condition $n \geq 1$ becomes $k \geq 0$. Option (A) would lose the equation for a_1.

My answer

Q28. [new]

Define $a_1 = 3$ and $a_n = a_{n-1} + 4$ for $n \geq 2$. Which explicit formula holds for all $n \geq 1$?

(A) $a_n = 4n + 3$

(B) $a_n = 3 \cdot 4^{n-1}$

(C) $a_n = 4n - 1$

(D) $a_n = 3n$

Answer: (C). $a_1 = 3, a_2 = 7, a_3 = 11$; if $a_k = 4k - 1$ then $a_{k+1} = 4k - 1 + 4 = 4(k + 1) - 1$.

My answer

Q29. Quiz 3 (Term 232)

Which of the following sequences satisfies the recurrence relation $a_n = 2a_{n-1} - 4a_{n-2}$ for all $n \geq 2$?

(A) $a_n = 2$

(B) $a_n = 2^n$

(C) $a_n = (-2)^n$

(D) None of these

Answer: (D). Substitute: $2 \neq 4 - 8$; 2^n gives $2^n - 2^n = 0 \neq 2^n$; $(-2)^n$ gives $(-2)^{n-2}(-4 - 4) = -2(-2)^n \neq (-2)^n$. (The original option $(1/2)^n$ also fails.)

My answer

Q30. [new]

Using $\sum_{j=0}^{n-1}(a + jd) = na + \dfrac{d(n-1)n}{2}$, the sum of the first 20 terms of the arithmetic sequence $5, 8, 11, 14, \ldots$ is:

(A) 670

(B) 630

(C) 740

(D) 1340

Answer: (A). $a = 5$, $d = 3$, $n = 20$: $20 \cdot 5 + \frac{3 \cdot 19 \cdot 20}{2} = 100 + 570 = 670$.

My answer

Q31. Quiz 3 Alkhulaifi

A collector starts with 10 cars and buys 2 new cars at the end of every month. She pays \$20 per month to maintain each car she owns during that month. How much does she pay in total for maintenance after 1 year (12 months)? [Hint: You may need to use the sum of an arithmetic sequence.]

(A) \$5280

(B) \$5040

(C) \$4600

(D) \$5480

Answer: (B). Month j ($j = 0, \ldots, 11$) she owns $10 + 2j$ cars: $\sum_{j=0}^{11}(10 + 2j) = 120 + 132 = 252$ car-months, and $252 \times 20 = 5040$.

My answer

Q32. [new]

A theatre has 15 rows. The first row has 20 seats and each row has 2 more seats than the row in front of it. How many seats are there in total?

(A) 480

(B) 500

(C) 540

(D) 510

Answer: (D). Arithmetic sum with $a = 20$, $d = 2$, $n = 15$: $15 \cdot 20 + \frac{2 \cdot 14 \cdot 15}{2} = 300 + 210 = 510$.

My answer

Q33. [new]

In the inductive proof of the closed form for an arithmetic sum, the inductive hypothesis is $\sum_{j=0}^{k-1}(a + jd) = ka + \frac{d(k-1)k}{2}$. Applying it to $(a + kd) + \sum_{j=0}^{k-1}(a + jd)$ gives:

(A) $(a + kd) + (k + 1)a + \dfrac{dk(k + 1)}{2}$

(B) $(a + kd) + ka + \dfrac{d(k - 1)k}{2}$

(C) $ka + \dfrac{d(k - 1)k}{2}$

(D) $(a + kd)\left(ka + \dfrac{d(k - 1)k}{2}\right)$

Answer: (B). Only the summation is replaced; the separated last term $a + kd$ stays. Simplifying gives $(k + 1)a + \frac{dk(k+1)}{2}$.

My answer

Q34. [new]

Using $\sum_{j=0}^{n-1} ar^j = \dfrac{a(r^n - 1)}{r - 1}$ for $r \neq 1$, the value of $2 + 6 + 18 + \cdots + 2 \cdot 3^9$ is:

(A) $3^9 - 1$

(B) $3^{10} - 1$

(C) $2(3^{10} - 1)$

(D) $\dfrac{3^{10} - 1}{2}$

Answer: (B). $a = 2$, $r = 3$, $n = 10$ terms: $\frac{2(3^{10}-1)}{2} = 3^{10} - 1 = 59048$.

My answer

Q35.

Quiz 3 Sec 3 (Term 232)

The sum of the first 40 terms in the series $3, 15, 75, 375, \ldots$ can be denoted as:

(A) $S = \sum_{i=0}^{40}(3 \cdot 5^{i-1})$

(B) $S = \sum_{i=0}^{39}(4 \cdot 5^{i-1})$

(C) $S = \sum_{i=1}^{40}(3 \cdot 5^{i-1})$

(D) $S = \sum_{i=2}^{41}(4 \cdot 5^{i-1})$

Answer: (C). The i-th term is $3 \cdot 5^{i-1}$ ($i = 1$ gives 3), and i runs over 40 values $1, \ldots, 40$.

My answer

Q36.

Quiz 3 Sec 3 (Term 232)

Sami earned SR100,000 during the first year of his job. After each year, he received a 10% raise. His total earnings from this job after spending 10 years is equal to:

(A) $1.1^9(100{,}000)$

(B) $1.1^{10}(100{,}000)$

(C) $\sum_{i=0}^{10}\left(1.1^i(100{,}000)\right)$

(D) $\sum_{i=0}^{9}\left(1.1^i(100{,}000)\right)$

Answer: (D). Year $i+1$ pays $100{,}000 \cdot 1.1^i$ for $i = 0, \ldots, 9$ (ten years); (A) and (B) are single-year salaries and (C) has 11 terms.

My answer

Q37.

[new]

In the inductive proof of $\sum_{j=0}^{n-1} ar^j = \dfrac{a(r^n - 1)}{r - 1}$ ($r \neq 1$, $n \geq 1$), the inductive hypothesis is $\sum_{j=0}^{k-1} ar^j = \dfrac{a(r^k - 1)}{r - 1}$. What must be proven?

(A) $\sum_{j=0}^{k} ar^j = \dfrac{a(r^{k+1} - 1)}{r - 1}$

(B) $\sum_{j=0}^{k+1} ar^j = \dfrac{a(r^{k+1}-1)}{r-1}$

(C) $\sum_{j=0}^{k} ar^j = \dfrac{a(r^{k}-1)}{r-1}$

(D) $\sum_{j=1}^{k} ar^j = \dfrac{a(r^{k+1}-1)}{r-1}$

Answer: (A). Replace n by $k+1$: the upper limit becomes $(k+1)-1=k$ and the exponent becomes $k+1$.

My answer

Q38.

Quiz 3 (Term 232)

The summation $\sum_{i=1}^{n}\sum_{j=1}^{n} j$ is equal to:

(A) n^2+n

(B) $(n+1)^2/2$

(C) $(n^2+n)/2$

(D) $(n^3+n^2)/2$

Answer: (D). The inner sum is $\frac{n(n+1)}{2}$ and does not depend on i; adding it n times gives $\frac{n^2(n+1)}{2} = \frac{n^3+n^2}{2}$.

My answer

Q39.

Quiz 3 Sec 3 (Term 232)

The value of the double summation $\displaystyle\sum_{i=0}^{10}\sum_{j=1}^{8}\left(2^j - j(3^i)\right)$ is equal to:

(A) $10(2^9-2) - \dfrac{36(3^{10}-1)}{2}$

(B) $11(2^9-2) - \dfrac{36(3^{11}-1)}{2}$

(C) $11(2^9) - \dfrac{36(3^{11}-1)}{2}$

(D) $10(2^9-1) - \dfrac{36(3^{10}-1)}{2}$

Answer: (B). $\sum_{j=1}^{8} 2^j = 2^9-2$ is added for 11 values of i; $\sum_{j=1}^{8} j = 36$ and $\sum_{i=0}^{10} 3^i = \frac{3^{11}-1}{2}$.

My answer

Q40. [new]

An inductive proof of the generalized De Morgan's law $\overline{A_1 \cap A_2 \cap \cdots \cap A_n} = \overline{A_1} \cup \overline{A_2} \cup \cdots \cup \overline{A_n}$ for $n \geq 2$ uses which inductive hypothesis?

(A) $\overline{A_1 \cap A_2} = \overline{A_1} \cup \overline{A_2}$

(B) $\overline{A_1 \cap \cdots \cap A_k} = \overline{A_1} \cup \cdots \cup \overline{A_k}$ for some $k \geq 2$

(C) $\overline{A_1 \cap \cdots \cap A_{k+1}} = \overline{A_1} \cup \cdots \cup \overline{A_{k+1}}$

(D) $\overline{A_1 \cup \cdots \cup A_k} = \overline{A_1} \cap \cdots \cap \overline{A_k}$

Answer: (B). The hypothesis is the statement for k sets; (A) is the base case and (C) is what must be proven.

My answer

Q41. [new]

In the inductive step of the generalized De Morgan's law for sets, let $B = A_1 \cap \cdots \cap A_k$. The equality $\overline{B \cap A_{k+1}} = \overline{B} \cup \overline{A_{k+1}}$ is justified by:

(A) the inductive hypothesis

(B) the distributive law

(C) De Morgan's law for two sets

(D) the definition of B

Answer: (C). B is a single set, so this is the two-set De Morgan's law; the inductive hypothesis is used afterwards to expand $\overline{B}$.

My answer

Q42. [new]

To prove $\neg(p_1 \vee p_2 \vee \cdots \vee p_n) \equiv \neg p_1 \wedge \neg p_2 \wedge \cdots \wedge \neg p_n$ for $n \geq 2$ by induction, the inductive step starts from $\neg(p_1 \vee \cdots \vee p_k \vee p_{k+1})$ and first rewrites it as:

(A) $\neg(p_1 \vee \cdots \vee p_k) \vee \neg p_{k+1}$

(B) $\neg(p_1 \vee \cdots \vee p_k) \wedge \neg p_{k+1}$

(C) $(\neg p_1 \wedge \cdots \wedge \neg p_k) \vee \neg p_{k+1}$

(D) $\neg p_1 \vee \cdots \vee \neg p_{k+1}$

Answer: (B). Treat $q = p_1 \vee \cdots \vee p_k$ as one proposition: $\neg(q \vee p_{k+1}) \equiv \neg q \wedge \neg p_{k+1}$; then apply the inductive hypothesis to $\neg q$.

My answer

Q43. [new]

The generalized distributive law $q \wedge (p_1 \vee p_2 \vee \cdots \vee p_n) \equiv (q \wedge p_1) \vee (q \wedge p_2) \vee \cdots \vee (q \wedge p_n)$ is proven by induction for $n \geq 2$. What is the base case?

(A) $q \wedge p_1 \equiv q \wedge p_1$

(B) $q \vee (p_1 \wedge p_2) \equiv (q \vee p_1) \wedge (q \vee p_2)$

(C) $q \wedge (p_1 \vee \cdots \vee p_k) \equiv (q \wedge p_1) \vee \cdots \vee (q \wedge p_k)$

(D) $q \wedge (p_1 \vee p_2) \equiv (q \wedge p_1) \vee (q \wedge p_2)$

Answer: (D). For $n = 2$ the statement is the ordinary distributive law of $\wedge$ over $\vee$.

My answer

Q44. [new]

The Fibonacci sequence is defined by $f_0 = 0$, $f_1 = 1$ and $f_n = f_{n-1} + f_{n-2}$ for $n \geq 2$. What is f_{10}?

(A) 34

(B) 89

(C) 55

(D) 144

Answer: (C). $0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55$, so $f_{10} = 55$.

My answer

Q45. [new]

Why is strong induction (rather than ordinary induction) the natural tool for proving $f_n \leq 2^n$ for the Fibonacci numbers?

(A) Because $f_{k+1} = f_k + f_{k-1}$, so bounds on both f_k and f_{k-1} are needed.

(B) Because the base case $f_0 \leq 2^0$ is false.

(C) Because 2^n grows faster than f_n.

(D) Because ordinary induction can only prove equalities.

Answer: (A). The inductive hypothesis of ordinary induction gives only $f_k \leq 2^k$; the recurrence also needs $f_{k-1} \leq 2^{k-1}$.

My answer

Q46. [new]

A strong induction proof shows that $f_n \leq 2^n$ for all $n \geq 0$ (Fibonacci, $f_0 = 0$, $f_1 = 1$). The inductive step, for $k \geq 1$, uses $f_{k+1} = f_k + f_{k-1}$. Which base cases must be proven directly?

(A) Only $f_0 \leq 2^0$

(B) Only $f_1 \leq 2^1$

(C) $f_0 \leq 2^0$, $f_1 \leq 2^1$ and $f_2 \leq 2^2$

(D) $f_0 \leq 2^0$ and $f_1 \leq 2^1$

Answer: (D). The step starts at $k = 1$ and uses $k - 1 \geq 0$, so it needs $n = 0$ and $n = 1$ already established; f_2 is then covered by the step.

My answer

Q47.

Quiz 3 Sec 3 (Term 232)

What distinguishes strong induction from regular mathematical induction?

(A) It requires proving the statement for multiple base cases.

(B) It allows assuming the statement is true for all values less than $n = k$.

(C) It involves proving the statement for all values greater than $n = k$.

(D) It uses a stronger base case.

Answer: (B). The inductive hypothesis covers all earlier values, not just one. Multiple base cases are common but not required (e.g. prime factorization uses one) (old key said A).

My answer

Q48.

Quiz 3 Sec 3 (Term 232)

When using strong induction, what is the role of the induction hypothesis?

(A) To assume the statement is true for all values less than $n = k$.

(B) To assume the statement is true for some $n = k$.

(C) To prove the statement for $n = k + 1$.

(D) To prove the statement for all values less than $n = k$.

Answer: (A). The hypothesis is an assumption about all values up to the current one; proving the next value is the job of the inductive step.

My answer

Q49.

Quiz 3 Sec 3 (Term 232)

What is the goal of strong induction?

(A) To prove the statement for a single base case.

(B) To establish the truth of the statement for a particular value of n.

(C) To strengthen the base case in a regular induction proof.

(D) To prove the statement for all values of n (in the given range).

Answer: (D). Like ordinary induction, strong induction concludes that the statement holds for every n in the range.

My answer

Q50.

Quiz 3 Sec 3 (Term 232)

In strong induction, what is the first step?

(A) Prove the statement for the base case.

(B) Assume the statement is true for some $n = k$.

(C) Assume the statement is true for all values less than $n = k$.

(D) Prove the statement for all values less than $n = k$.

Answer: (A). Every induction proof starts by proving the base case(s) directly.

My answer

Q51.

Quiz 3 Sec 3 (Term 232)

How does strong induction differ from regular induction in terms of the induction step?

(A) Strong induction does not have an induction step.

(B) Strong induction requires proving the statement for multiple consecutive values of n.

(C) Strong induction uses a stronger inductive step.

(D) Strong induction allows assuming the statement is true for all values less than $n = k$.

Answer: (D). What is stronger is the hypothesis, not the step. The implication $(P(a) \wedge \cdots \wedge P(k)) \rightarrow P(k+1)$ is actually easier to prove than $P(k) \rightarrow P(k+1)$ (old key said C).

My answer

Q52.

[new]

The inductive step of a strong induction proof shows, for $k \geq 1$, that if $S(0) \wedge S(1) \wedge \cdots \wedge S(k)$ is true then $S(k+1)$ is true. Which of the following may NOT be used as part of the inductive hypothesis?

(A) $S(0)$

(B) $S(k-1)$

(C) $S(k)$

(D) $S(k+1)$

Answer: (D). $S(k+1)$ is what must be proven; every $S(j)$ with $0 \leq j \leq k$ (including $S(k-1)$, since $k \geq 1$) may be assumed.

My answer

Q53. [new]

Juice is sold in 3-packs and 4-packs. Let $P(n)$: "n cans can be bought using only 3-packs and 4-packs". A strong induction proof that $P(n)$ holds for all $n \geq 6$ adds one 3-pack to $k-2$ cans in the inductive step. Which base cases are needed?

(A) $P(6)$ only

(B) $P(6)$ and $P(7)$

(C) $P(6)$, $P(7)$ and $P(8)$

(D) $P(6)$, $P(7)$, $P(8)$ and $P(9)$

Answer: (C). The step for $k+1$ uses $P(k-2)$ and needs $k-2 \geq 6$, i.e. $k \geq 8$; so $P(6), P(7), P(8)$ must be proven directly ($6 = 3+3$, $7 = 3+4$, $8 = 4+4$).

My answer

Q54. [new]

In the same juice proof (3-packs and 4-packs, $n \geq 6$), the inductive step for $k \geq 8$ argues that $k-2$ cans can be bought. Why is the condition $k \geq 8$ needed?

(A) So that $k-2 \geq 6$ lies in the range covered by the inductive hypothesis.

(B) So that $k+1$ is divisible by 3.

(C) So that $k-2$ is even.

(D) Because $P(5)$ is false.

Answer: (A). The inductive hypothesis covers $6 \leq j \leq k$; $k-2$ is in that range exactly when $k \geq 8$.

My answer

Q55. [new]

Juice is now sold in 4-packs and 5-packs, and $P(n)$ is proven for all $n \geq 12$ by strong induction, where the inductive step adds one 4-pack to $k-3$ cans. Which base cases must be proven directly?

(A) $P(12)$, $P(13)$, $P(14)$ and $P(15)$

(B) $P(12)$, $P(13)$ and $P(14)$

(C) $P(12)$ and $P(13)$

(D) $P(12)$ only

Answer: (A). Adding a 4-pack requires 4 consecutive base cases: $12 = 4+4+4$, $13 = 4+4+5$, $14 = 4+5+5$, $15 = 5+5+5$; then $k-3 \geq 12$ for $k \geq 15$.

My answer

Q56.

Quiz 3 Alkhulaifi

For $n \geq 8$, consider the statement $P(n)$: "the integer n can be written as a sum of 3's and 5's." (For example $11 = 3+3+5$.) If we want to prove the statement using strong induction, then a sufficient set of basis steps (the base case) is:

(A) Prove $P(8)$ is true

(B) Prove $P(8)$, $P(9)$ and $P(10)$ are true

(C) Prove $P(1), P(2), \ldots, P(7)$ are true

(D) Prove $P(8)$ and $P(9)$ are true

Answer: (B). The step writes $k+1 = (k-2)+3$, which needs $k-2 \geq 8$, i.e. $k \geq 10$; so $8 = 3+5$, $9 = 3+3+3$, $10 = 5+5$ must be checked. (The original option "$P(8), \ldots, P(12)$" is also sufficient, so it was dropped.)

My answer

Q57.

Quiz 3 (Term 232)

Which of the following sets of stamps can make all postage values ≥ 20 cents?

(A) 5 and 8 cents

(B) 4 and 7 cents

(C) 5 and 7 cents

(D) 3 and 9 cents

Answer: (B). With 4 and 7 every amount ≥ 18 works ($18 = 4+7+7$, $19 = 4 \cdot 3 + 7$, $20 = 4 \cdot 5$, $21 = 7 \cdot 3$, then add 4's). 27 fails for $5, 8$; 23 fails for $5, 7$; $3, 9$ give only multiples of 3.

My answer

Q58.

HW 5

Let $P(n)$: "postage of n cents can be formed using just 4-cent and 7-cent stamps". A strong induction proof for $n \geq 18$ proves $P(18), P(19), P(20), P(21)$ directly. In the inductive step for $k \geq 21$, which part of the inductive hypothesis is used to prove $P(k+1)$?

(A) $P(k)$

(B) $P(k-6)$

(C) $P(18)$

(D) $P(k-3)$

Answer: (D). $k+1 = (k-3)+4$; since $k \geq 21$, $18 \leq k-3 \leq k$, so $P(k-3)$ holds and one more 4-cent stamp gives $k+1$.

My answer

Q59. HW 5

Which amounts of postage (in cents) can be formed using just 3-cent and 10-cent stamps?

(A) Every amount ≥ 18, and no smaller amount

(B) $3, 6, 9, 10, 12, 13, 15, 16$, and every amount ≥ 18

(C) Every multiple of 3, and every amount ≥ 10

(D) $3, 6, 9, 10, 12, 13, 15, 16, 17$, and every amount ≥ 18

Answer: (B). Below 18 only $3a + 10b$ with small a, b work ($17 = 3a + 10b$ is impossible); $18 = 3 \cdot 6$, $19 = 3 \cdot 3 + 10$, $20 = 10 \cdot 2$, then add 3-cent stamps.

My answer

Q60. [new]

In the strong induction proof that every integer $n \geq 2$ is a product of primes, the inductive step writes a composite $k + 1 = a \cdot b$. What must be shown about a and b to apply the inductive hypothesis?

(A) a and b are both prime.

(B) $2 \leq a \leq k$ and $2 \leq b \leq k$.

(C) $a = k$ and $b = 1$.

(D) a and b are both even.

Answer: (B). The hypothesis covers every integer from 2 through k; since $a, b \geq 2$, each factor is at most $(k + 1)/2 < k + 1$.

My answer

Q61. [new]

The well-ordering principle states that:

(A) every nonempty set of integers has a smallest element.

(B) every nonempty subset of the nonnegative integers has a largest element.

(C) every nonempty subset of the nonnegative integers has a smallest element.

(D) every subset of the real numbers has a smallest element.

Answer: (C). Restricting to nonnegative integers is essential: $\mathbb{Z}$ itself has no smallest element.

My answer

Q62. [new]

Which of the following nonempty sets has NO smallest element?

(A) $\{n \in \mathbb{Z} : n \geq 100\}$

(B) The set of prime numbers

(C) $\{n^2 : n \in \mathbb{Z}^+\}$

(D) $\{x \in \mathbb{Q} : x > 0\}$

Answer: (D). For any positive rational x, $x/2$ is a smaller positive rational. The other sets are nonempty subsets of the nonnegative integers, so well-ordering gives them a least element.

My answer

Q63. [new]

In the well-ordering proof of the Division Algorithm, $r = n - dq$ is the smallest element of $S = \{n - dq' : q' \in \mathbb{Z},\ n - dq' \geq 0\}$. To show $r < d$, one assumes $r \geq d$ and derives a contradiction using which element of S?

(A) $r + d = n - d(q - 1)$

(B) $r - d = n - d(q + 1)$

(C) $r - 1$

(D) n

Answer: (B). If $r \geq d$ then $r - d = n - d(q + 1) \geq 0$ is in S and is smaller than r, contradicting the minimality of r.

My answer

Q64. [new]

Define $a_1 = 1$, $a_2 = 2$ and $a_n = a_{n-1} + 2a_{n-2}$ for $n \geq 3$. Which explicit formula can be proven by strong induction?

(A) $a_n = n$

(B) $a_n = \dfrac{n^2 - n + 2}{2}$

(C) $a_n = 2^{n-1}$

(D) $a_n = n^2 - 2n + 2$

Answer: (C). All four formulas give $a_1 = 1, a_2 = 2$, but $a_3 = 4$ and $a_4 = 8$ rule out (A) (3), (B) ($a_4 = 7$) and (D) ($a_3 = 5$). Step: $a_{k+1} = 2^{k-1} + 2 \cdot 2^{k-2} = 2^k$.

My answer

Q65. [new]

Define $b_0 = 2$, $b_1 = 3$ and $b_n = 3b_{n-1} - 2b_{n-2}$ for $n \geq 2$. Which explicit formula holds for all $n \geq 0$?

(A) $b_n = 2^n + 1$

(B) $b_n = n + 2$

(C) $b_n = 2^{n+1} - n$

(D) $b_n = n^2 + 2$

Answer: (A). $b_2 = 9 - 4 = 5$, $b_3 = 15 - 6 = 9$; and $3(2^k + 1) - 2(2^{k-1} + 1) = 2^{k+1} + 1$. Each distractor matches b_0, b_1 but gives $b_2 \neq 5$.

My answer

Q66. [new]

Every positive integer can be written as a sum of distinct nonnegative integer powers of 2 (proved by strong induction). How is 45 written?

(A) $2^5 + 2^3 + 2^1 + 2^0$

(B) $2^5 + 2^3 + 2^2 + 2^0$

(C) $2^5 + 2^4 - 2^1 - 2^0$

(D) $2^4 + 2^4 + 2^3 + 2^2 + 2^0$

Answer: (B). $32 + 8 + 4 + 1 = 45$ ($45 = 101101_2$). (A) is 43, (C) uses subtraction, (D) repeats 2^4.

My answer

Q67. [new]

A sequence is given by a_1, a_2, a_3 and, for $n \geq 4$, a recurrence expressing a_n in terms of a_{n-1} and a_{n-3}. To prove an upper bound on a_n by strong induction (step for $k \geq 3$ proving the bound for a_{k+1}), how many base cases are needed?

(A) 1

(B) 2

(C) 3

(D) 4

Answer: (C). The step for a_{k+1} uses a_{k-2}, which is in the range only if $k - 2 \geq 1$, i.e. $k \geq 3$; so a_1, a_2, a_3 must be checked directly.

My answer

Q68. [new]

Which of the following is a recursive definition of the factorial function on the nonnegative integers?

(A) $0! = 0$; $n! = n \cdot (n-1)!$ for $n \geq 1$

(B) $0! = 1$; $n! = n \cdot (n-1)!$ for $n \geq 1$

(C) $n! = 1 \cdot 2 \cdots n$

(D) $1! = 1$; $n! = n + (n-1)!$ for $n \geq 2$

Answer: (B). A basis value $0! = 1$ plus the rule $n! = n \cdot (n-1)!$; (A) makes every value 0 and (C) is not recursive.

My answer

Q69. [new]

A function on the nonnegative integers is defined by $f(0) = 3$ and $f(n) = 2f(n-1) + 1$ for $n \geq 1$. What is $f(3)$?

(A) 15

(B) 25

(C) 31

(D) 63

Answer: (C). $f(1) = 7$, $f(2) = 15$, $f(3) = 31$.

My answer

Q70. HW 5

Let $f(0) = f(1) = 1$ and $f(n+1) = f(n) - f(n-1)$ for $n = 1, 2, \ldots$. What is $f(5)$?

(A) -2

(B) -1

(C) 1

(D) 0

Answer: (D). $f(2) = 0$, $f(3) = -1$, $f(4) = -1$, $f(5) = f(4) - f(3) = -1 - (-1) = 0$ (old key said -2).

My answer

Q71. HW 5

Let $f(0) = f(1) = 1$ and $f(n+1) = f(n) \cdot f(n-1)$ for $n = 1, 2, \ldots$. What is $f(5)$?

(A) 1

(B) 5

(C) 8

(D) 0

Answer: (A). $f(2) = 1 \cdot 1 = 1$ and every later value is again $1 \cdot 1 = 1$.

My answer

Q72. [new]

A function on the nonnegative integers is defined by $f(0) = 0$ and $f(n) = f(n-1) + n^2$ for $n \geq 1$. What is $f(4)$?

(A) 16

(B) 30

(C) 25

(D) 55

Answer: (B). $f(1) = 1$, $f(2) = 5$, $f(3) = 14$, $f(4) = 30$.

My answer

Q73. [new]

Which recursive definition (on the nonnegative integers) defines the function $f(n) = 2^n$?

(A) $f(0) = 0$; $f(n) = 2f(n-1)$ for $n \geq 1$

(B) $f(0) = 1$; $f(n) = f(n-1) + 2$ for $n \geq 1$

(C) $f(0) = 2$; $f(n) = 2f(n-1)$ for $n \geq 1$

(D) $f(0) = 1$; $f(n) = 2f(n-1)$ for $n \geq 1$

Answer: (D). $2^0 = 1$ and $2^n = 2 \cdot 2^{n-1}$. (A) gives all zeros, (B) gives $2n + 1$, (C) gives 2^{n+1}.

My answer

Q74. HW 5

Which is a recursive definition of the set S of odd positive integers?

(A) Basis: $1 \in S$. Rule: if $x \in S$ then $x + 2 \in S$.

(B) Basis: $1 \in S$. Rule: if $x \in S$ then $x + 1 \in S$.

(C) Basis: $0 \in S$. Rule: if $x \in S$ then $x + 2 \in S$.

(D) Basis: $1 \in S$. Rule: if $x \in S$ then $2x + 1 \in S$.

Answer: (A). Generates $1, 3, 5, \ldots$. (B) gives all positive integers, (C) the even nonnegatives, (D) only $1, 3, 7, 15, \ldots$.

My answer

Q75. HW 5

Which is a recursive definition of the set S of positive integer powers of 3, i.e. $\{3, 9, 27, \ldots\}$?

(A) Basis: $1 \in S$. Rule: if $x \in S$ then $3x \in S$.

(B) Basis: $3 \in S$. Rule: if $x \in S$ then $x + 3 \in S$.

(C) Basis: $3 \in S$. Rule: if $x \in S$ then $3x \in S$.

(D) Basis: $3 \in S$. Rule: if $x \in S$ then $x^3 \in S$.

Answer: (C). (A) also puts $3^0 = 1$ in S, which is not a positive integer power (old key used basis 1); (B) gives all positive multiples of 3; (D) gives $3, 27, 3^9, \ldots$ only.

My answer

Q76. [new]

In a recursive definition of a set, which component states that an element belongs to the set only if it is in the basis or can be built by applying the recursive rules finitely many times?

(A) The basis

(B) The recursive rule

(C) The exclusion statement

(D) The inductive hypothesis

Answer: (C). The exclusion statement rules out everything else; it is often left implicit.

My answer

Q77. [new]

Properly nested parentheses are defined by: Basis: `()` is properly nested. Rules: if x and y are properly nested then (1) `(`x`)` and (2) xy are properly nested. Which string is properly nested?

(A) `(())(`

(B) `())(()`

(C) `(()())`

(D) `((())`

Answer: (C). `(()())` $=$ `(`x`)` with $x =$ `()()`. (A) and (B) have a prefix with more `)` than `(`; (D) is unbalanced.

My answer

Q78. [new]

Using the rules of the previous question, what was the last rule applied in constructing `(()())(())`?

(A) Rule (1): `(`x`)`

(B) Rule (2): xy

(C) The basis

(D) It cannot be constructed.

Answer: (B). It is xy with $x =$ `(()())` and $y =$ `(())`; the first `(` is matched before the end, so it is not of the form `(`x`)`.

My answer

Q79. [new]

Using the same rules, what was the last rule applied in constructing `(((())(()))`?

(A) Rule (1): `(`x`)`

(B) Rule (2): xy

(C) The basis

(D) It cannot be constructed.

Answer: (A). The first `(` matches the last `)`, so it is `(`x`)` with $x =$ `(()())(())`.

My answer

Q80. Quiz 3 Alkhulaifi

Let S be the set of strings over the alphabet $\{a, b\}$ defined recursively as follows. Base case: $\lambda \in S$ and $a \in S$. Recursive rule: if $x \in S$ then $xb \in S$ and $xba \in S$. Then the set of all strings of length at most 3 in S is

(A) $\{\lambda, ab, ba, bb, aba, abb, bab, bba\}$

(B) $\{\lambda, a, b, ab, ba, bb, aba, abb, bab, bba, bbb\}$

(C) $\{\lambda, a, b, ab, ba, bb, aba, aaa, abb, bab, bba, bbb\}$

(D) $\{\lambda, a, b, ab, ba, bb, aba, bba, bbb\}$

Answer: (B). From λ: b, ba; from a: ab, aba; from b: bb, bba; from ab: abb; from ba: bab; from bb: bbb. aaa can never be built.

My answer

Q81. [new]

The set $\{0,1\}^*$ of binary strings is defined by: basis $\lambda \in \{0,1\}^*$; rule: if $x \in \{0,1\}^*$ then $x0$ and $x1$ are in $\{0,1\}^*$. How many applications of the recursive rule are needed to construct the string 1101?

(A) 3

(B) 5

(C) 1

(D) 4

Answer: (D). $\lambda \to 1 \to 11 \to 110 \to 1101$: one application per character.

My answer

Q82. [new]

A perfect binary tree is built from a single vertex by repeatedly taking two copies of the current tree and joining their roots to a new root. How many vertices does the tree obtained after 3 applications of the recursive rule have?

(A) 7

(B) 15

(C) 8

(D) 16

Answer: (B). $1 \to 3 \to 7 \to 15$, since each step gives $2v + 1$ vertices.

My answer

Q83. [new]

Let $v(T)$ be the number of vertices of a perfect binary tree T. Which is a correct recursive definition of v?

(A) $v(T) = 1$ for the single vertex; $v(T') = 2v(T) + 1$ if T' is built from two copies of T.

(B) $v(T) = 0$ for the single vertex; $v(T') = 2v(T)$.

(C) $v(T) = 1$ for the single vertex; $v(T') = v(T) + 2$.

(D) $v(T) = 1$ for the single vertex; $v(T') = 2v(T)$.

Answer: (A). T' has two copies of T plus the new root.

My answer

Q84. [new]

The height of a perfect binary tree is defined by $h(T) = 0$ for the single vertex and $h(T') = h(T) + 1$ when T' is built from two copies of T. What is the height of the perfect binary tree with 31 vertices?

(A) 3

(B) 5

(C) 4

(D) 15

Answer: (C). Vertices $1, 3, 7, 15, 31$ correspond to heights $0, 1, 2, 3, 4$ (a tree of height h has $2^{h+1} - 1$ vertices).

My answer

Q85. [new]

Full binary trees are defined by: a single vertex is a full binary tree; if T_1 and T_2 are full binary trees, a new root with left subtree T_1 and right subtree T_2 is a full binary tree (left/right order matters). How many different full binary trees have exactly 7 vertices?

(A) 3

(B) 4

(C) 14

(D) 5

Answer: (D). Subtree sizes $(1,5),(3,3),(5,1)$ give $1\cdot 2+1\cdot 1+2\cdot 1=5$ trees (there are 1 tree with 1 or 3 vertices and 2 with 5).

My answer

Q86. [new]

For full binary trees (as in the previous question), which is a correct recursive definition of the height h?

(A) $h(T)=0$ for a single vertex; $h(T)=h(T_1)+h(T_2)+1$.

(B) $h(T)=0$ for a single vertex; $h(T)=1+\max(h(T_1),h(T_2))$.

(C) $h(T)=1$ for a single vertex; $h(T)=1+\min(h(T_1),h(T_2))$.

(D) $h(T)=0$ for a single vertex; $h(T)=2+\max(h(T_1),h(T_2))$.

Answer: (B). The longest downward path goes through one edge from the root into the taller subtree.

My answer

Q87. [new]

Which recursive definition gives exactly the set of binary palindromes?

(A) Basis: $\lambda, 0, 1$. Rule: if x is in the set then $0x0$ and $1x1$ are in the set.

(B) Basis: λ. Rule: if x is in the set then $0x0$ and $1x1$ are in the set.

(C) Basis: $0, 1$. Rule: if x is in the set then $x0$ and $x1$ are in the set.

(D) Basis: $\lambda, 0, 1$. Rule: if x is in the set then $0x1$ and $1x0$ are in the set.

Answer: (A). Wrapping a palindrome with the same symbol on both ends keeps it a palindrome; the basis covers even and odd lengths. (B) misses odd-length palindromes; (C) gives all nonempty strings.

My answer

Q88. [new]

A subset S of the integers is defined by: basis $1 \in S$; rule: if $x \in S$ then $x+3 \in S$ and $x-3 \in S$. Which are the elements of S with absolute value less than 10?

(A) $1, 4, 7$

(B) $-8, -5, -2, 1, 4, 7$

(C) $-9, -6, -3, 0, 3, 6, 9$

(D) $-7, -4, -1, 1, 4, 7$

Answer: (B). $S = \{1 + 3m : m \in \mathbb{Z}\}$; those with $|x| < 10$ are $-8, -5, -2, 1, 4, 7$.

My answer

Q89. [new]

Which is a recursive definition of the set E of all even integers (positive, negative and zero)?

(A) Basis: $0 \in E$. Rule: if $x \in E$ then $x + 2 \in E$.

(B) Basis: $2 \in E$. Rule: if $x \in E$ then $x + 2 \in E$.

(C) Basis: $0 \in E$. Rule: if $x \in E$ then $2x \in E$.

(D) Basis: $0 \in E$. Rule: if $x \in E$ then $x + 2 \in E$ and $x - 2 \in E$.

Answer: (D). Both $x + 2$ and $x - 2$ are needed to reach the negative even integers. (A) gives only $0, 2, 4, \ldots$; (B) only $2, 4, 6, \ldots$; (C) only 0.

My answer

Q90. HW 5

Let $S \subseteq \mathbb{Z} \times \mathbb{Z}$ be defined recursively by: basis $(0, 0) \in S$; rule: if $(a, b) \in S$ then $(a + 2, b + 3) \in S$ and $(a + 3, b + 2) \in S$. Which elements are produced by exactly the fourth application of the recursive rule?

(A) $(6, 9), (7, 8), (8, 7), (9, 6)$

(B) $(10, 15), (11, 14), (12, 13), (13, 12), (14, 11), (15, 10)$

(C) $(8, 12), (9, 11), (10, 10), (11, 9), (12, 8)$

(D) $(8, 12), (10, 10), (12, 8)$

Answer: (C). After j applications, $a + b = 5j$ with $a = 2j + i$, $i = 0, \ldots, j$; for $j = 4$ this gives five pairs. (The old homework list jumped from the third level straight to the fifth and omitted these.)

My answer

Part B: Written problems

Q91.

Quiz 4 Alrasasi

Use mathematical induction to prove that for every integer $n \geq 1$,

$$\sum_{j=1}^{n} j(j+1) = \frac{n(n+1)(n+2)}{3}.$$

Answer:

Let $P(n)$ be the statement $\sum_{j=1}^{n} j(j+1) = \frac{n(n+1)(n+2)}{3}$. By induction on n.

Base case ($n = 1$). Left side: $1 \cdot 2 = 2$. Right side: $\frac{1 \cdot 2 \cdot 3}{3} = 2$. So $P(1)$ is true.

Inductive step. Suppose that for some integer $k \geq 1$, $\sum_{j=1}^{k} j(j+1) = \frac{k(k+1)(k+2)}{3}$ (inductive hypothesis). We show $\sum_{j=1}^{k+1} j(j+1) = \frac{(k+1)(k+2)(k+3)}{3}$.

$$\begin{aligned}
\sum_{j=1}^{k+1} j(j+1) &= \sum_{j=1}^{k} j(j+1) + (k+1)(k+2) && \text{(separate the last term)} \\
&= \frac{k(k+1)(k+2)}{3} + (k+1)(k+2) && \text{(inductive hypothesis)} \\
&= \frac{k(k+1)(k+2) + 3(k+1)(k+2)}{3} \\
&= \frac{(k+1)(k+2)(k+3)}{3}.
\end{aligned}$$

So $P(k+1)$ is true. By the principle of mathematical induction, $P(n)$ holds for every integer $n \geq 1$. ■

My answer

Q92.

HW 5

Let $P(n)$ be the statement $1^3 + 2^3 + \cdots + n^3 = \left(\frac{n(n+1)}{2}\right)^2$ for the positive integer n. (a) What is $P(1)$, and show it is true. (b) What is the inductive hypothesis? (c) What must be proven in the inductive step? (d) Complete the inductive step. (e) Explain why these steps show the formula is true for all positive integers n.

Answer:

(a) $P(1)$: $1^3 = \left(\frac{1\cdot 2}{2}\right)^2$. Both sides equal 1, so $P(1)$ is true.

(b) For an arbitrary integer $k \geq 1$: $1^3 + 2^3 + \cdots + k^3 = \left(\frac{k(k+1)}{2}\right)^2$.

(c) For every $k \geq 1$: if $P(k)$ is true, then $1^3 + \cdots + k^3 + (k+1)^3 = \left(\frac{(k+1)(k+2)}{2}\right)^2$.

(d)

$$\begin{aligned} 1^3 + \cdots + k^3 + (k+1)^3 &= \frac{k^2(k+1)^2}{4} + (k+1)^3 && \text{(inductive hypothesis)} \\ &= \frac{(k+1)^2\left(k^2 + 4(k+1)\right)}{4} = \frac{(k+1)^2(k^2+4k+4)}{4} \\ &= \frac{(k+1)^2(k+2)^2}{4} = \left(\frac{(k+1)(k+2)}{2}\right)^2. \end{aligned}$$

So $P(k) \to P(k+1)$ for every $k \geq 1$.

(e) $P(1)$ is true, and $P(1) \to P(2)$, $P(2) \to P(3)$, and so on; by the principle of mathematical induction $P(n)$ is true for every positive integer n. ■

My answer

Q93.

HW 5

Prove by mathematical induction that for every positive integer n,

$$1 \cdot 2 \cdot 3 + 2 \cdot 3 \cdot 4 + \cdots + n(n+1)(n+2) = \frac{n(n+1)(n+2)(n+3)}{4}.$$

Answer:

By induction on n.

Base case ($n = 1$). Left side: $1 \cdot 2 \cdot 3 = 6$. Right side: $\frac{1\cdot 2\cdot 3\cdot 4}{4} = 6$. True.

Inductive step. Assume for some $k \geq 1$ that $\sum_{j=1}^{k} j(j+1)(j+2) = \frac{k(k+1)(k+2)(k+3)}{4}$. Then

$$\begin{aligned} \sum_{j=1}^{k+1} j(j+1)(j+2) &= \frac{k(k+1)(k+2)(k+3)}{4} + (k+1)(k+2)(k+3) && \text{(inductive hypothesis)} \\ &= \frac{(k+1)(k+2)(k+3)\,(k+4)}{4}, \end{aligned}$$

which is the formula with $n = k + 1$: $\frac{(k+1)((k+1)+1)((k+1)+2)((k+1)+3)}{4}$.

Hence the statement holds for $k + 1$, and by induction it holds for every positive integer n. ■

My answer

Q94. HW 5

Prove that $n^2 - 7n + 12$ is nonnegative whenever n is an integer with $n \geq 3$.

Answer:

Let $P(n)$: $n^2 - 7n + 12 \geq 0$. By induction on n, for $n \geq 3$.

Base case ($n = 3$). $9 - 21 + 12 = 0 \geq 0$. True.

Inductive step. Suppose that for some integer $k \geq 3$, $k^2 - 7k + 12 \geq 0$. Then

$$\begin{aligned}(k+1)^2 - 7(k+1) + 12 &= k^2 + 2k + 1 - 7k - 7 + 12 \\ &= (k^2 - 7k + 12) + (2k - 6).\end{aligned}$$

By the inductive hypothesis $k^2 - 7k + 12 \geq 0$, and since $k \geq 3$, $2k - 6 \geq 0$. A sum of two nonnegative numbers is nonnegative, so $(k+1)^2 - 7(k+1) + 12 \geq 0$, i.e. $P(k+1)$ holds.

By induction, $n^2 - 7n + 12 \geq 0$ for every integer $n \geq 3$. (Note that $2k - 6 \geq 0$ follows directly from $k \geq 3$; no second induction is needed.) ■

My answer

Q95. Quiz 4 Alenzi

Prove that for any positive integer n, 4 evenly divides $11^n - 7^n$.

Answer:

By induction on n.

Base case ($n = 1$). $11^1 - 7^1 = 4 = 4 \cdot 1$, which is divisible by 4.

Inductive step. Assume for some integer $k \geq 1$ that $4 \mid 11^k - 7^k$, i.e. $11^k - 7^k = 4m$ for some integer m. Then

$$\begin{aligned} 11^{k+1} - 7^{k+1} &= 11 \cdot 11^k - 11 \cdot 7^k + 11 \cdot 7^k - 7 \cdot 7^k \\ &= 11(11^k - 7^k) + (11 - 7)\,7^k \\ &= 11(4m) + 4 \cdot 7^k = 4\,(11m + 7^k). \end{aligned}$$

Since $11m + 7^k$ is an integer, $4 \mid 11^{k+1} - 7^{k+1}$.

By induction, 4 evenly divides $11^n - 7^n$ for every positive integer n. ■

My answer

Q96.

HW 5

Prove that if $A_1, A_2, \ldots, A_n$ and B are sets ($n \geq 1$), then

$$(A_1 - B) \cup (A_2 - B) \cup \cdots \cup (A_n - B) = (A_1 \cup A_2 \cup \cdots \cup A_n) - B.$$

Answer:

By induction on n.

Base case ($n = 1$). Both sides are $A_1 - B$.

Two-set fact. For any sets X, Y, B: $(X - B) \cup (Y - B) = (X \cap \overline{B}) \cup (Y \cap \overline{B}) = (X \cup Y) \cap \overline{B} = (X \cup Y) - B$, by the distributive law.

Inductive step. Assume for some $k \geq 1$ that $(A_1 - B) \cup \cdots \cup (A_k - B) = (A_1 \cup \cdots \cup A_k) - B$. Let $X = A_1 \cup \cdots \cup A_k$. Then

$$\begin{aligned}
&(A_1 - B) \cup \cdots \cup (A_k - B) \cup (A_{k+1} - B) \\
&\quad = \big((A_1 \cup \cdots \cup A_k) - B\big) \cup (A_{k+1} - B) && \text{(inductive hypothesis)} \\
&\quad = (X - B) \cup (A_{k+1} - B) \\
&\quad = (X \cup A_{k+1}) - B && \text{(two-set fact)} \\
&\quad = (A_1 \cup \cdots \cup A_k \cup A_{k+1}) - B.
\end{aligned}$$

So the identity holds for $k + 1$, and by induction it holds for every $n \geq 1$. ■

My answer

Q97. Quiz 4 Alenzi

Define the sequence $\{a_n\}$ by $a_1 = a_2 = a_3 = 1$ and $a_n = a_{n-1} + a_{n-2} + a_{n-3}$ for $n \geq 4$. Prove that $a_n < 2^n$ for all $n \geq 1$.

Answer:

By strong induction on n.

Base cases. $a_1 = 1 < 2$, $a_2 = 1 < 4$, $a_3 = 1 < 8$.

Inductive step. Let $k \geq 3$ and assume that $a_j < 2^j$ for every j with $1 \leq j \leq k$. We prove $a_{k+1} < 2^{k+1}$. Since $k+1 \geq 4$, the recurrence gives $a_{k+1} = a_k + a_{k-1} + a_{k-2}$. Because $k \geq 3$, the indices $k, k-1, k-2$ all lie in the range 1 through k, so by the inductive hypothesis

$$a_{k+1} < 2^k + 2^{k-1} + 2^{k-2} = 2^{k-2}(4+2+1) = 7 \cdot 2^{k-2} < 8 \cdot 2^{k-2} = 2^{k+1}.$$

Thus $a_{k+1} < 2^{k+1}$. Three base cases are needed because the step uses the three preceding terms. By strong induction, $a_n < 2^n$ for all $n \geq 1$. ■

My answer

Q98. HW 5

Let $P(n)$ be the statement that a postage of n cents can be formed using just 4-cent stamps and 7-cent stamps. Prove, using strong induction, that $P(n)$ is true for $n \geq 18$:
(a) show $P(18), P(19), P(20), P(21)$; (b) state the inductive hypothesis; (c) state what must be proven; (d) complete the inductive step for $k \geq 21$; (e) explain why the steps prove the statement.

Answer:

(a) $18 = 4 \cdot 1 + 7 \cdot 2$, $19 = 4 \cdot 3 + 7 \cdot 1$, $20 = 4 \cdot 5$, $21 = 7 \cdot 3$. So the four base cases hold.

(b) For some integer $k \geq 21$: $P(j)$ is true for every integer j with $18 \leq j \leq k$.

(c) $P(k+1)$: $k+1$ cents can be formed from 4-cent and 7-cent stamps.

(d) Since $k \geq 21$, we have $18 \leq k-3 \leq k$, so by the inductive hypothesis $k-3 = 4x + 7y$ for some nonnegative integers x, y. Adding one 4-cent stamp: $k+1 = 4(x+1) + 7y$. So $P(k+1)$ is true.

(e) The base cases cover 18–21, and the inductive step shows that once all values from 18 to k are formable (for $k \geq 21$), so is $k+1$. Thus $P(22), P(23), \ldots$ follow in turn, and by the principle of strong induction $P(n)$ holds for all $n \geq 18$. Four base cases are needed because the step reaches back 3 values ($k+1-4 = k-3$). ■

My answer

Q99. HW 5

Suppose you begin with a pile of n stones and split it into n piles of one stone each by successively splitting a pile into two smaller piles. Each time you split a pile into piles of r and s stones, you compute the product rs. Use strong induction to show that, no matter how the piles are split, the sum of all the products computed equals $\dfrac{n(n-1)}{2}$.

Answer:

Let $P(n)$: every way of splitting a pile of n stones into n single stones produces products summing to $\frac{n(n-1)}{2}$. By strong induction on $n \geq 1$.

Base case ($n = 1$). No split is performed, the sum is 0, and $\frac{1 \cdot 0}{2} = 0$.

Inductive step. Let $k \geq 1$ and assume $P(j)$ for every $1 \leq j \leq k$. Consider a pile of $k+1 \geq 2$ stones. The first split produces piles of i and $k+1-i$ stones for some $1 \leq i \leq k$, contributing the product $i(k+1-i)$. After that, the two piles are split independently. Since $1 \leq i \leq k$ and $1 \leq k+1-i \leq k$, the inductive hypothesis applies to both piles, so the remaining products sum to $\frac{i(i-1)}{2} + \frac{(k+1-i)(k-i)}{2}$.

Writing $m = k+1-i$ (so $i+m = k+1$), the total is

$$im + \frac{i(i-1)}{2} + \frac{m(m-1)}{2} = \frac{2im + i^2 - i + m^2 - m}{2} = \frac{(i+m)^2 - (i+m)}{2}$$
$$= \frac{(k+1)^2 - (k+1)}{2} = \frac{(k+1)k}{2}.$$

This does not depend on i or on the later splits, so $P(k+1)$ holds. By strong induction, $P(n)$ is true for all $n \geq 1$. ■

My answer

Q100. HW 5

Let $S \subseteq \mathbb{Z}\times\mathbb{Z}$ be defined recursively by: basis $(0,0) \in S$; recursive step: if $(a,b) \in S$ then $(a+2, b+3) \in S$ and $(a+3, b+2) \in S$. (a) Use induction on the number of applications of the recursive step to show that $5 \mid a+b$ for every $(a,b) \in S$. (b) Is the converse true, i.e. is every pair of nonnegative integers with $5 \mid a+b$ in S?

Answer:

(a) Let $Q(n)$: every pair obtained from $(0,0)$ by exactly n applications of the recursive step satisfies $a+b = 5n$. By the exclusion statement, every element of S is obtained by some number $n \geq 0$ of applications, so it suffices to prove $Q(n)$ for all $n \geq 0$.

Base case ($n = 0$). The only such pair is $(0,0)$ and $0+0 = 0 = 5\cdot 0$.

Inductive step. Assume $Q(k)$ for some $k \geq 0$. A pair obtained with $k+1$ applications is $(a+2, b+3)$ or $(a+3, b+2)$, where (a,b) was obtained with k applications, so $a+b = 5k$ by the inductive hypothesis. In either case the new sum is $a+b+5 = 5k+5 = 5(k+1)$. So $Q(k+1)$ holds.

By induction $Q(n)$ holds for all $n \geq 0$; hence every $(a,b) \in S$ has $a+b = 5n$ for some n, so $5 \mid a+b$. ■

(b) No. For example $(5,0)$ has $5 \mid 5+0$, but a pair with $a+b = 5$ needs exactly one application, which only produces $(2,3)$ and $(3,2)$. So $(5,0) \notin S$. (In fact, after n applications a ranges over $2n, 2n+1, \ldots, 3n$, so $S = \{(a,b) : a+b = 5n,\ 2n \leq a \leq 3n,\ n \geq 0\}$.)

My answer